2.0 Reference Designs

2.1 Containers with multiple sensors

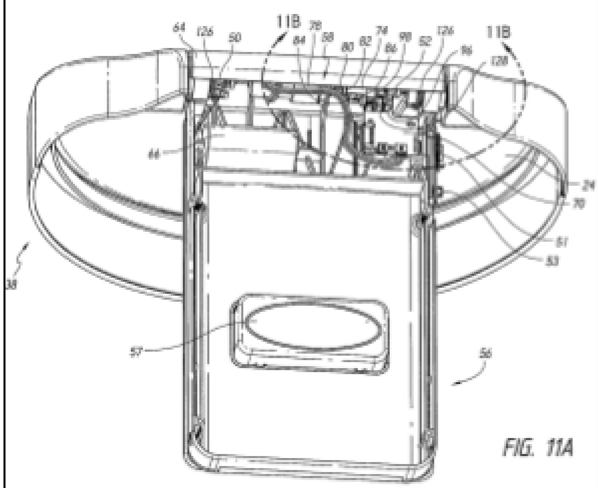

Figure 5. *A trashcan with an automated lid, improving ease of use.*

This reference design prioritizes accessibility. It describes a trashcan with a sensor assembly, allowing for automated movement of the lid [1]. This inspires the goal for vending machine flaps to be operable with minimal force required from the user. However, it fails to uphold the integrity of the anti-theft features of vending machines, as it stays open as long and as wide as needed by the user.

2.2 Long-handled Variable-angle dustpan

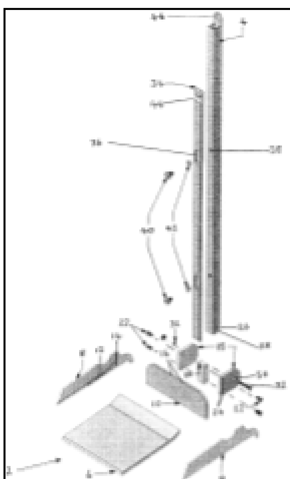

Figure 6. *A long-handled dustpan patent.*

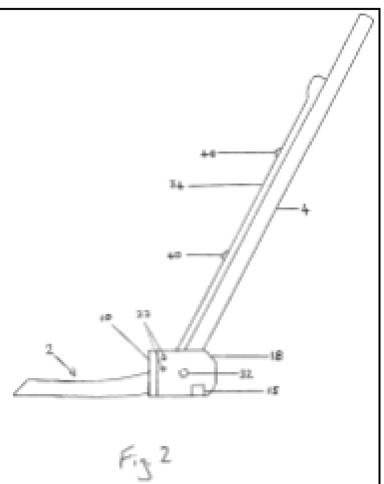

Figure 7. *A variable-angle handle dustpan patent.*

Retrieving products from vending machines and using traditional dustpans both require excessive bending, causing lower back strain. This dustpan prioritises ergonomics and accessibility, and is designed to reduce strain by increasing the height as shown in Figures 6 and 7 [2]. This design inspired the goal of increasing the current product retrieval height. However, this design fails to address the vending machine opening's purpose of retrieving, not disposing of products.

2.3 Product Discharge and Delivery System

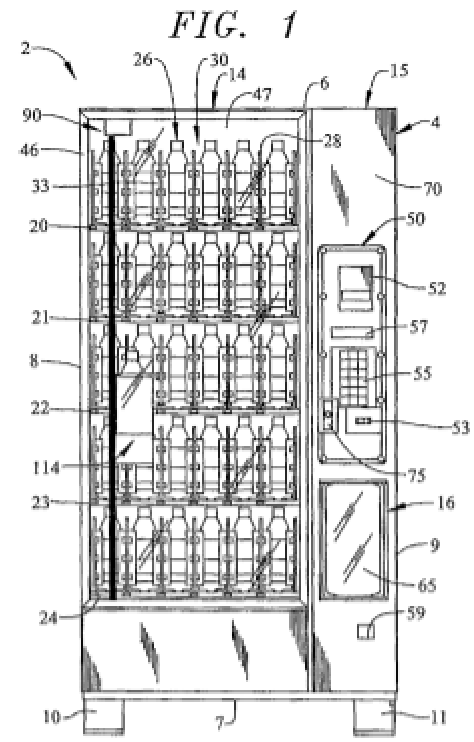

Figure 8. *A beverage vending machine with a carriage system and alternate access port.*

This reference design takes an alternative approach to incorporate an anti-pilfer device [3], as seen in Figure 8. It has a system that carries the product to an easily accessible chamber without a flap, making it easier to use. However, it is not implementable on the current machine and would require replacing it. Additionally, it is beverage exclusive and therefore does not meet the needs of the primary stakeholders.